

Zainab Tairu

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Skills

- **Programming & Tools:** Python, SQL, Git, GitHub
- **Machine Learning:** Transformer Models, Deep Learning, Agentic AI, Predictive Modelling
- **Natural Language Processing:** RAG, LLMs, Prompt Engineering, Rasa, LangChain, spaCy, HuggingFace
- **Data Engineering:** ETL Pipelines, Data Integration, Feature Engineering, Automation
- **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, FastAPI, Streamlit, Gradio
- **Databases:** Neo4j (Graph Databases), DynamoDB
- **Cloud & MLOps:** AWS (EC2, S3, Lambda)
- **Mathematical Foundations:** Linear Algebra, Calculus, Probability, Statistics

Experience

NNPC Upstream Investment Management, Lagos

Oct 2025 – Present

Intern

- Designed an ETL pipeline automating daily production reports from field operators, improving data accessibility and reporting efficiency.
- Explored ML applications for predictive maintenance and reservoir characterization using well log and seismic data.

CoMMA Lab, Texas (Remote)

Jul 2025 – Present

NLP Research Intern

- Engineered NLP algorithms to model polymer properties and co-authored a survey exploring the intersection of LLMs and polymer modeling.

The Innoverse, South Africa (Remote)

Oct 2024 – Jul 2025

NLP Intern

- Built an intelligent chatbot integrating RAG and LLMs to enhance response accuracy and personalization.
- Developed knowledge graph structures and trained multilingual text-to-speech models for English and Zulu.

Projects

RFE Risk Analyzer

Jul 2025

- Created a USCIS-style risk analyzer using DeepSeek LLM and Ollama to simulate legal adjudicator feedback on EB-1A petitions.

LET ME COOK

Nov 2024

- Developed a Nigerian recipe dataset and app for efficient pantry-based recipe recommendations. [View here.](#)

Thermostable Lipase Engineering

Jun 2025

- Applied transformer-based protein models to predict stability-enhancing mutations in *Bacillus* sp. lipase.

Malaria Blood Smear Classifier

Jun 2025

- Built a CNN algorithm achieving 99% accuracy for malaria diagnosis from blood smear images.

Publications

Papers and Preprints

Multimodal Model for Polymer Property Prediction. In preparation (Expected 2026)

Poster Presentations

Hot and Sustainable: AI-Guided Design of Thermostable Lipases. Deep Learning Indaba 2025

Dual AI Architectures for Accessible Medication Management: OCR-LLM vs. VLM. Deep Learning Indaba 2025

Reservoir Characterization with Machine Learning. SPE OAU Hackathon 2025

Leadership & Activities

Deep Learning Lead, DSN OAU Coordinated speaker sessions on ethical AI and advanced deep learning research	Jan 2025–Present
Public Relations Officer, Space Club OAU Managed public communications and organized STEM outreach initiatives	Jan 2025–Oct 2025
Deep Learning Tutor,DSN OAU Taught CNN fundamentals and guided students through model development with scikit learn.	May 2025–Jun 2025
Data Science Tutor, Star Girls Africa Taught 100+ students Python programming and supervised their first machine learning models	Jun 2024–Aug 2024
Member, SPE OAU Participated in hackathons and projects applying ML in petroleum engineering	Jan 2022–Present